



Agriassist – smart AI based Farming assistant

Course : software project managemen

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Faculty : Prof.L.Jerart Julus

Assessment : Course based project

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1.INTRODUCTION:

AgriAssist – Smart AI-Based Farming Assistant is a web-based application designed to help farmers make informed agricultural decisions using modern technologies. The agriculture sector faces several challenges, including poor soil quality, crop diseases, improper crop selection, and unpredictable weather conditions. This system aims to address these issues by providing intelligent and technology-driven solutions to improve productivity and efficiency in farming.

This project uses Artificial Intelligence, Machine Learning, Image Processing, and Web Technologies to deliver smart solutions for farmers. The system analyzes soil using image processing techniques and predicts soil pH and nutrient levels accurately. Based on soil nutrients such as Nitrogen, Phosphorus, and Potassium, along with environmental factors like temperature, humidity, and rainfall, the system recommends suitable crops for cultivation. The application also detects plant diseases by analyzing leaf images using advanced AI and image processing techniques. It provides pesticide suggestions and weather forecast information to help farmers plan their agricultural activities effectively. The system includes an AI chatbot called AgriBot to answer farmer queries and provide instant assistance. It also supports multiple languages to ensure easy usability for farmers from different regions. The main aim of this project is to improve crop productivity and promote efficient and smart farming practices.

2.LITERATURE SURVEY:

From the literature survey, it is observed that many researchers have developed systems for plant disease detection using machine learning and deep learning techniques. Image processing and CNN models are widely used for accurate prediction. Some systems also use IoT and weather data for smart farming. However, most existing systems focus only on a single feature and lack integration. Therefore, our proposed AgriAssist system combines multiple features such as soil analysis, disease detection, crop recommendation, weather forecasting, and chatbot support into a single intelligent platform.

	Year of publishing		
Research paper name		Idea	Limitations

Plant disease detection using deep learning	2016	Used CNN to detect plant diseases from leaf images	Works only for disease detection, no soil analysis, crop recommendation and weather support.
Crop recommendation	2018	Used ML algorithm to suggest	No image based soil analysis, no disease
system using machine learning		crop based on soil data	detection, no real time weather integration.
Smart farming using IoT and ML	2019	Uses sensors to collect soil data	Requires expensive IoT devices, not accessible to small farmers and no image based detection.

3.PROBLEM STATEMENT:

Agriculture plays a vital role in the economy, but farmers face many challenges such as improper crop selection, poor soil fertility, plant diseases, lack of knowledge about fertilizers and pesticides, and unpredictable weather conditions. Many farmers do not have access to proper soil testing facilities, so they are unable to know the soil pH level and nutrient content such as Nitrogen, Phosphorus, and Potassium. Due to this, they may choose unsuitable crops, which leads to low productivity and financial loss.

Another major problem is the late detection of plant diseases, which spreads quickly and damages crops. Farmers also find it difficult to get proper guidance about pesticide usage and crop management. In addition, language barriers and lack of technical knowledge make it difficult for farmers to use modern agricultural tools. Therefore, there is a need for a smart, easy-to-use system that can analyze soil, recommend suitable crops, detect plant diseases, provide pesticide guidance, and offer weather information and farmer support through an AI chatbot. This system will help farmers make better decisions, improve crop yield, reduce crop loss, and support modern smart farming practices

AI + Image Processing + Weather Data.

4.METHODOLOGY:

The AgriAssist system collects data such as soil images, soil nutrients (N, P, K), pH value, temperature, humidity, and rainfall. The collected data is preprocessed, and image processing is performed using OpenCV to analyze soil and plant images. Machine Learning algorithms are used to recommend suitable crops and detect plant diseases. The backend processes the data and communicates with the trained models and database. Finally, the web application displays the results such as soil analysis, crop recommendation, disease detection, pesticide guide, and chatbot assistance in a user-friendly interface.

The application takes user input and database support to provide outputs like crop type, soil parameters, and crop price recommendations.

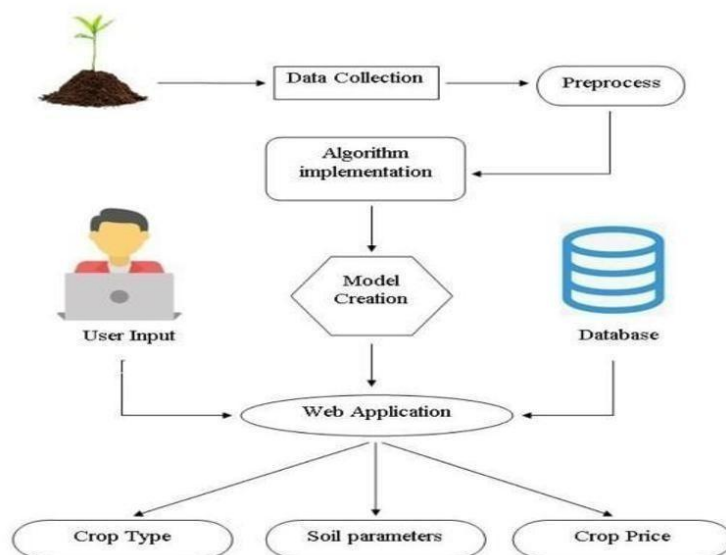


Figure 1 shows workflow of agriassist system showing data collection, preprocessing, model creation, and web application for crop and soil analysis

5.ALGORITHMS:

we used used four algorithms , they are:

5.1 . CNN – Convolutional Neural Network

CNN is used for plant disease detection. The user uploads a leaf image. The CNN model analyzes the image. It compares the image with trained dataset images. Then it predicts whether the plant is healthy or diseased and identifies the disease name.

5.2 . Regression Algorithm

Regression is used for soil pH prediction. The system analyzes soil data such as soil texture, nutrients, etc. Regression algorithm predicts the pH value of the soil. The pH value helps to decide which crop is suitable.

5.3 . Random Forest Algorithm

Random Forest is used for crop recommendation. The system takes input like: Nitrogen, Phosphorus, Potassium, pH value, Temperature Rainfall, Humidity, Random Forest analyzes this data. Then it recommends the best crop to grow.

5.4 . NLP – Natural Language Processing

NLP is used in the AI chatbot. Farmer can ask questions in text. NLP understands the question. Chatbot gives appropriate answer.

6.RESULT AND DISCUSSION:

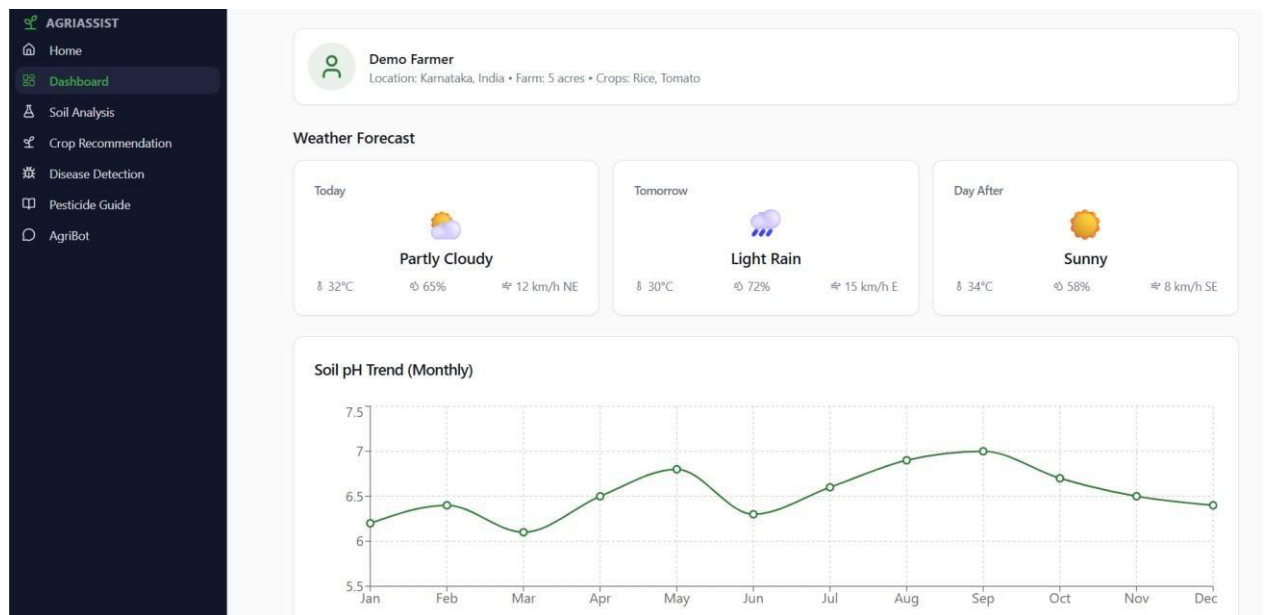


Figure 2 shows the agri assist dashboard interface, where the user can view weather forecasts, soil analysis, and farming insights. it displays temperature, humidity, and upcoming weather conditions along with soil fertility.



Figure 3 shows the agri assist welcome page, which introduces the agricultural assistant system. The page also includes contact details, helpline information, and a chatbot support option for farmers.



Figure 4 shows the agriassist home page, where users can select their preferred language such as english, hindi etc.

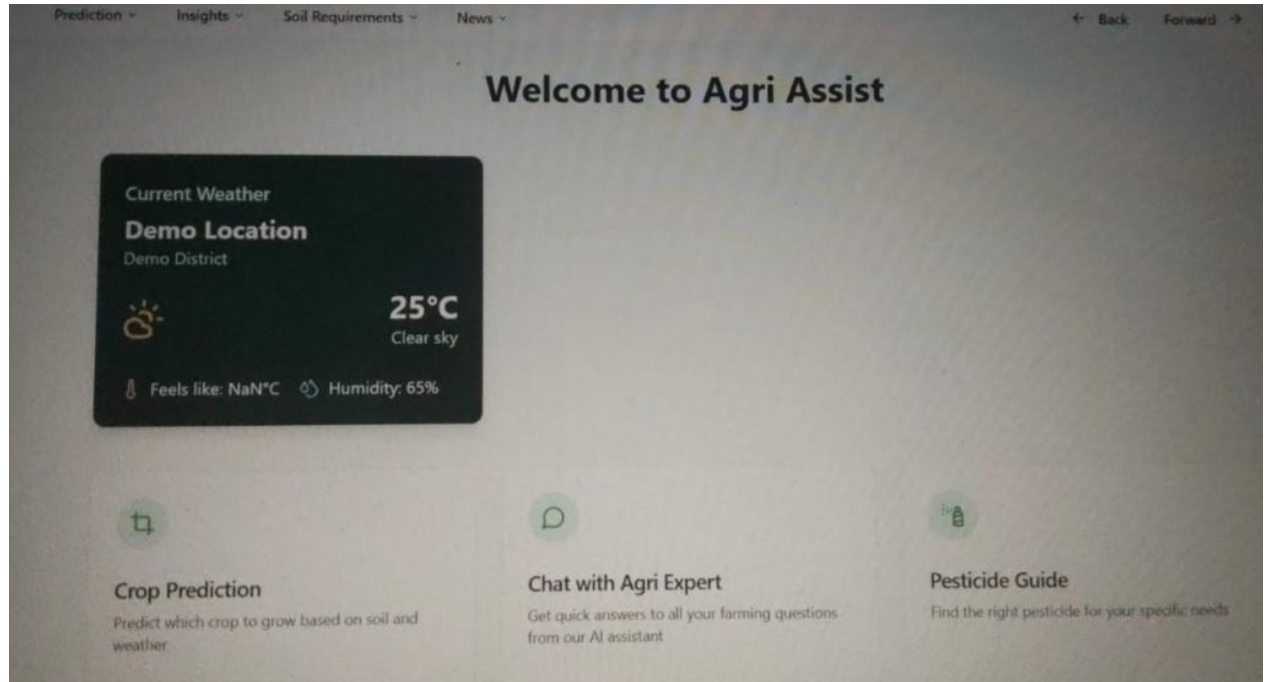


Figure 5 shows the main dashboard, where users can access features like crop prediction, fertilizer guide, pesticide guide, soil requirements, crop rotation and chat with an AI expert. system also fetch location of user to provide crop recommendation

Crop Recommendation
home
Contact
About

Crop Recommendation System ?

Nitrogen

Phosphorus

Potassium

Temperature

Humidity

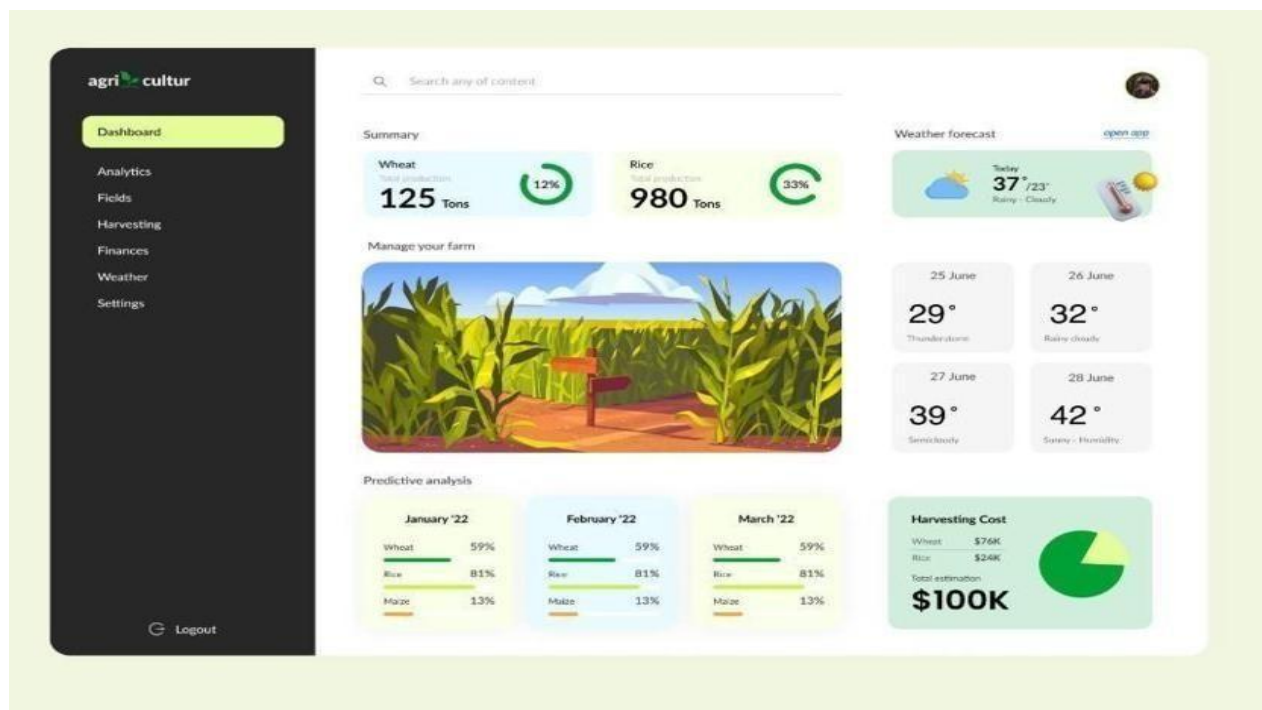
pH

Rainfall

{% if result %}

Recommend Crop for cultivation is:
{{ result }}

Figure 6 shows the crop recommendation interface, system analyse the inputs of soil nutrient and find ph value to recommend suitable crop.



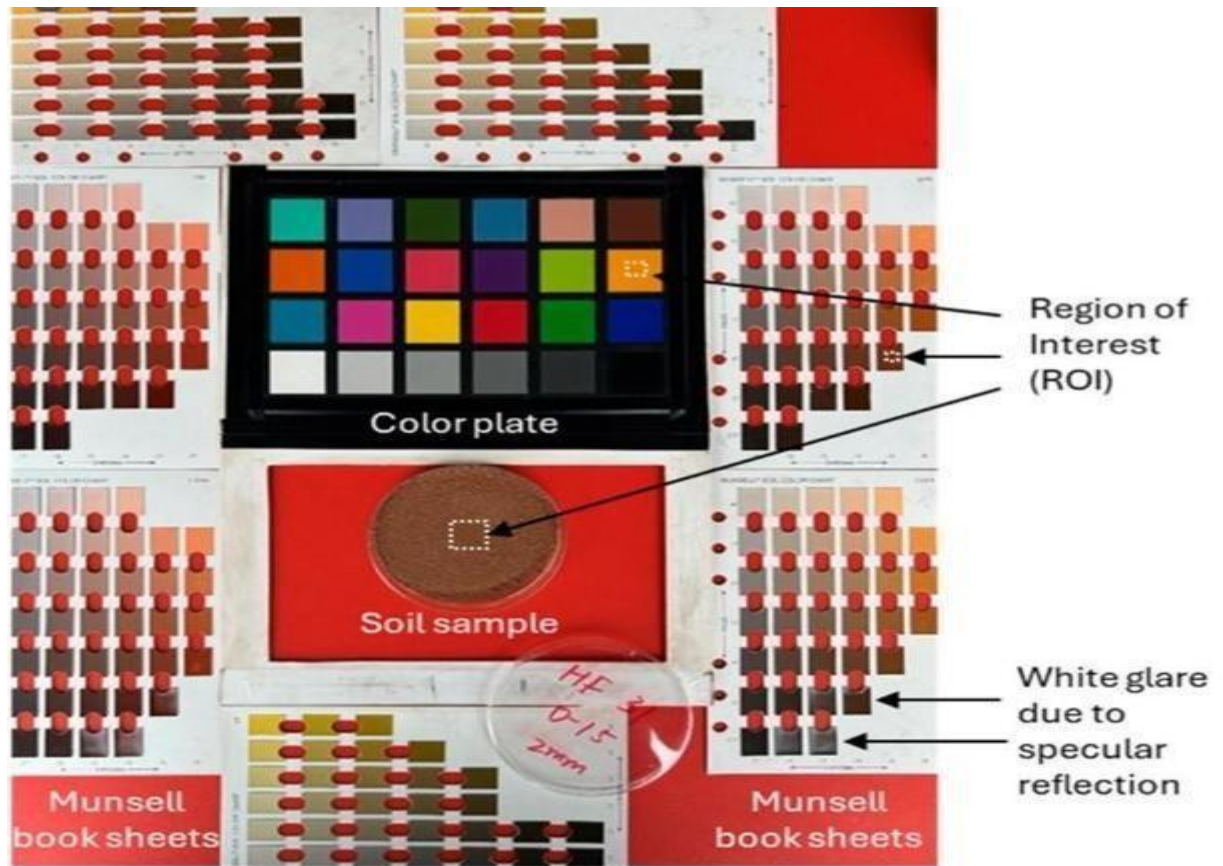


Figure 8 shows a soil sample placed on a color plate(Munsell color chart),where a region of interest(ROI) is selected to capture the soil color accurately.System compares the soil color with standard color sheets to estimate soil properties.

The AgriAssist system outputs useful information such as recommended crop types with suitability scores, soil health details including pH and NPK values, plant disease detection results with symptoms and treatment suggestions, suitable pesticides, weather forecasts, and chatbot responses in the selected language. It solves agricultural problems by collecting user inputs (soil data or images), preprocessing them, and using OpenCV for image analysis and Machine Learning algorithms to compare the data with trained datasets for accurate predictions. The backend processes these results and sends them to the web application, where farmers can easily understand and use the recommendations to improve crop selection, manage soil effectively, detect diseases early, and ultimately increase productivity and reduce losses.

7.REFERENCES :

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